VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH UNIVERSITY OF TECHNOLOGY

FACULTY OF CHEMICAL ENGINEERING

MECHANICAL PROCESSING

Laboratory Of Unit Operations (CH3015)

CONTINUOUS DISTILLATION

CLASS CC03 — GROUP 9 — SEMESTER 252

Instructor: Dr. Phạm Hoàng Huy Phước Lợi

Student's Name	**Student's ID**
Đinh Chấn Phong	2352902
Đinh Nguyễn Chính Nghĩa	2352804
Trần Hữu Đạt	2352245

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1 ABSTRACT

1.1 Experimental purpose

The objective of the experiment is to understand and apply basic concepts in distillation of binary mixtures. Continuous distillation of ethanol water mixture will be investigated:

- Determine the overall efficiency of the column
- Investigate effect of reflux ratio on the product composition
- Investigate effect of location of feed flow on the product composition

1.2 Experimental methods

This lab is mainly designed to separate binary components including ethanol and water using distillation with 5 stages batch column.

2 EXPERIMENTAL THEORY

The separation dynamics are most heavily dependent on the vapor pressure of the species in the column. The more volatile component is also known as the light key, the less volatile component is also known as the heavy key. For a two-component separation:

$$\alpha = \frac{K_L}{K_H} \tag{1}$$

$$K_L = \frac{P_L}{P} \tag{2}$$

$$K_H = \frac{P_H}{P} \tag{3}$$

All symbols are defined in the nomenclature section at the end of the text. By Raoult's law, this means that relative volatility can be described in terms of mole fractions in the liquid and vapor phases.

$$\alpha = \frac{y_L/x_L}{y_H/x_H} = \frac{y_L(1-x_L)}{x_L(1-y_L)} \tag{4}$$

This can be rearranged to give:

$$y_L = \frac{\alpha x_L}{1 + x_L(\alpha - 1)} \tag{5}$$

The traditional approach to modeling distillation processes is known as the McCabe-Thiele approach. This model requires making the following assumptions:

- That the relative volatility is constant over the temperature range in the column.
- That the components have equal and constant molar enthalpy.
- That enthalpy changes and heat of mixing are negligible.
- That the pressure is uniform within the column.

These assumptions allow equations to be derived relating flow rates to molar compositions in each stage of the column. The column is divided into the vapor phase, and the rectification section, in which the less volatile component is selectively condensed. In each section, the liquid and vapor compositions can be related to flow rates.

Rectification section:

$$V_{n+1}y_{n+1} = L_n x_n + D x_D \tag{6}$$

$$y_{n+1} = \frac{L_n}{V_{n+1}} x_n + \frac{D}{V_{n+1}} x_D \tag{7}$$

where the stages are numbered consecutively, beginning with the top stage.

$$\frac{L}{V} = \frac{R}{R+1} \tag{8}$$

$$y_n = \frac{R}{R+1} x_n + \frac{1}{R+1} x_D \tag{9}$$

Stripping section:

$$y_n = \frac{L_n}{V_n} x_n - \frac{B}{V_n} x_B \tag{10}$$

$$V_B = \frac{V}{B} \tag{11}$$

$$\frac{L}{V} = \frac{V+B}{V} = \frac{V_B + 1}{V_B} \tag{12}$$

$$y_n = \frac{V_B + 1}{V_B} x_n - \frac{1}{V_B} x_B \tag{13}$$

Given equilibrium data for the components of the distillation, these equations can be used to model the stages graphically, as shown in Figure **??**.

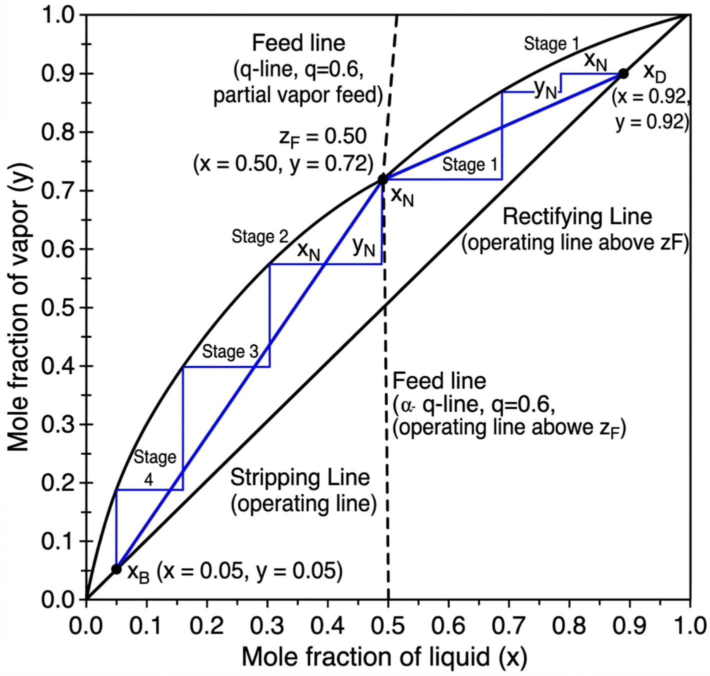


Figure 1: Sample McCabe-Thiele diagram. Each step indicates a theoretical stage going to complete equilibrium. The calculations are shown for the rectification section only.

Because true vapor-liquid equilibrium is unlikely to be achieved on each stage, this method must be adapted in order to properly model the distillation system. A widely used method for this is known as Murphree Efficiency:

$$E_M = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}} \tag{14}$$

Where y^* used in this equation is derived from experimentally measured liquid-phase composition values. The Murphree Efficiency can be used to alter the equilibrium line in the McCabe-Thiele graph. This adapted graph can be used to more accurately predict the separation under different conditions of feed composition and reflux ratios.

3 EXPERIMENT EQUIPMENT AND METHOD

3.1 Experiment Equipment

The experiment uses a continuous distillation column with the following components:

- 5-stage batch column.
- Reboiler with heating element.
- Condenser and reflux splitter.
- Feed pump and preheater.
- Temperature sensors at various stages.

3.2 Experiment Method

The experiment was conducted according to the following steps:

1. Prepare the binary mixture of ethanol and water with a known concentration.
2. Start the heating system and wait for the column to reach steady state.
3. Adjust the feed flow rate and reflux ratio according to the experimental matrix.
4. Measure temperatures and collect samples from the distillate and bottoms for concentration analysis.
5. Repeat for different reflux ratios and feed locations.

4 EXPERIMENTAL RESULTS

4.1 Raw Data

Table 1: Raw experimental results

No	**Loc. Feed**	**F (ml/min)**	**D (ml/min)**	**Reflux** L_o	**Conc. (%)** Feed	Dist.	**Temp. (°C)** t_F	t_W	t_D	t_{Lo}
1	4	150	44	7	10	45	57	93	38	88
2	4	150	59	10	10	55	58	96	38	83
3	4	150	28	15	10	45	58	93	38	79
4	2	150	36	10	10	50	55	96	37	78
5	5	150	46	10	10	50	64	96	38	77

4.2 Data Processing

Table 2: Raw results processing (mol fractions)

No	**Loc. Feed**	**F**[1] **(ml/min)**	**D (ml/min)**	**Reflux** L_o	x_F	x_D	x_W
1	4	150.0	44.0	39.48	0.0416	0.242	0.032
2	4	150.0	59.0	56.40	0.0416	0.323	0.016
3	4	150.0	28.0	84.60	0.0416	0.242	0.032
4	2	150.0	36.0	56.40	0.0416	0.281	0.016
5	5	150.0	46.0	56.40	0.0416	0.281	0.016

Table 3: Calculation Results

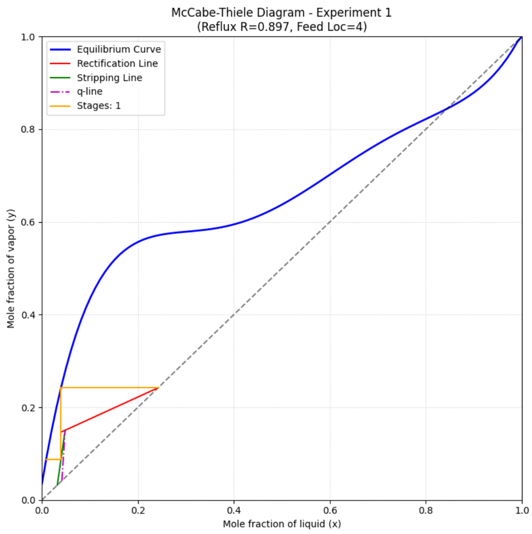
No	**Loc.**	**R**	t_F	H_F	H_{GF}	H_{LF}	q
1	4	0.90	57	235	2983	308	1.065
2	4	0.96	58	238	2983	307	1.063
3	4	3.02	58	238	2963	287	1.063
4	2	1.57	55	225	3003	327	1.069
5	5	1.23	64	262	3005	330	1.052

Table 4: Operating lines and efficiency results

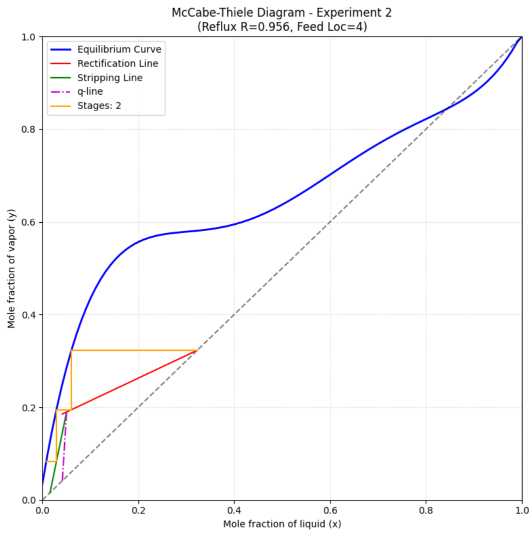
No	**Input line**	**Rectification line**	n_{LT}	**Ratio**^{2}	E_o
1	$y = 16.38x - 0.64$	$y = 0.47x + 0.13$	1	0.2	0.20
2	$y = 16.87x - 0.66$	$y = 0.49x + 0.17$	2	0.4	0.40
3	$y = 16.87x - 0.66$	$y = 0.75x + 0.06$	1	0.2	0.20
4	$y = 15.49x - 0.60$	$y = 0.61x + 0.11$	2	0.4	0.40
5	$y = 20.23x - 0.80$	$y = 0.55x + 0.13$	2	0.4	0.40

4.3 Graphical Analysis

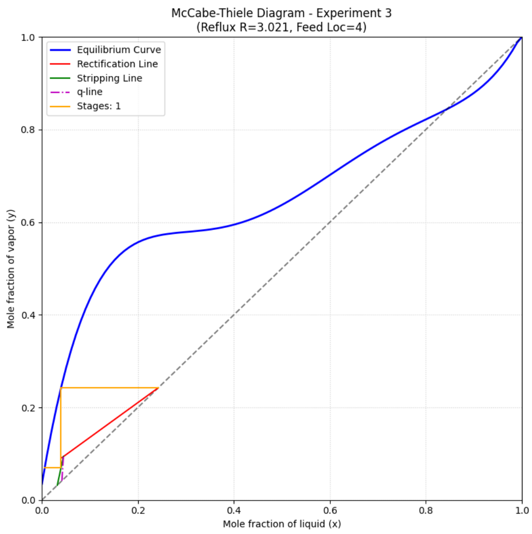
Schematic calculation of theoretical trays was performed for each experiment using the Python calculation suite. The resulting McCabe-Thiele diagrams are shown in Figure **??**.



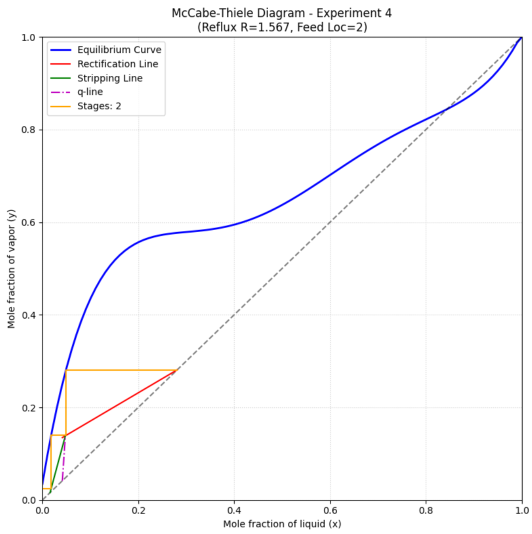
(a) Experiment 1



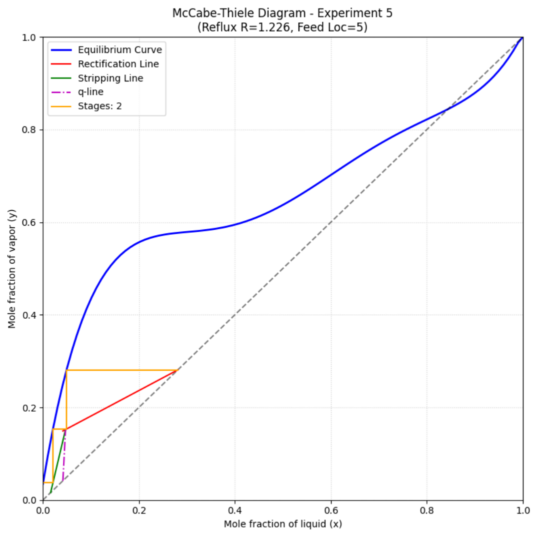
(b) Experiment 2



(c) Experiment 3



(d) Experiment 4



(e) Experiment 5

Figure 2: McCabe-Thiele diagrams for all five experiments.

5 DISCUSSION

5.1 Effect of reflux ratio on product purity and column performance

Based on experiments 1, 2, and 3, it is evident that as the reflux ratio (R) increases, the purity of the product (distillate) also increases. This phenomenon can be explained by the energy balance in the column. With a constant heat supply, the total molar vapor flow (V) remains relatively constant.

Since $V = L + D$, an increase in the reflux flow (L) results in a decrease in the distillate product flow (D). For the distillation column, the heat supplied to the reboiler serves to separate the phases. When the product flow D decreases, the specific heat supplied per mole of product increases, leading to better separation and higher purity in both top and bottom products.

However, it must be noted that general performance does not allow for a complete assessment of the economic viability of the equipment. While an increased reflux flow improves the separation efficiency of each tray—potentially reducing the total number of trays required and lowering initial capital costs—it also increases the individual heat supply requirement per unit of product. This energy cost can account for a significant percentage of the total operating costs, and must be balanced against the desired product purity.

5.2 Effect of feed location on purity and performance

The location of the feed tray significantly influences the concentration profile within the column. As the vapor and liquid streams pass through each tray, their concentrations change based on the exchange capacity of the tray.

Inputting the feed at trays near the bottom increases the number of rectification stages above the feed, generally increasing the distillate purity. Conversely, feeding near the top reduces the rectification stages. However, if the physical feed position deviates significantly from the optimal theoretical feed tray, the overall efficiency of the trays decreases. Experimental results show that the input tray position does not directly affect the individual tray efficiency, but rather how closely the column operates to its theoretical potential.

5.3 Stable operating conditions

Stable operation is achieved when the heat exchangers ensure consistent heat supply and the flow meters show steady values. Each tray maintains a distinct boiling temperature, which decreases progressively from the bottom (reboiler) to the top (condenser). To maintain stability, the following are required:

- Proper operation of the heat exchangers.
- Stable flow rates for feed, reflux, and distillate.

- Consistent temperature measurements across the trays (higher trays indicate lower temperatures).

5.4 Error analysis and mitigation

Sources of error during the experiment include:

- Unstable flow rates due to valves not being fully open or mechanical fluctuations.
- **Rotameter instability:** The observation that ”marbles” (floats) are dropped or oscillate, requiring constant monitoring of the flow.
- Measurement inaccuracies in alcohol concentration using hydrometers (”alcohol by incorrect design”).
- Timing discrepancies when reading multiple sensors simultaneously.
- Data rounding and plotting inaccuracies during the calculation phase.

To mitigate these errors, measurements should be taken simultaneously, flow rates must be monitored and adjusted continuously, and calculations should be verified using standardized tables and software where applicable.

6 APPENDIX

6.1 Physical Properties and Constants

The following data for Ethanol and Water was used in the calculations, distinguishing between molar and mass-based units:

Ethanol (C_2H_5OH):

- Molecular weight: $M_r = 46$ g/mol
- Specific weight: $\rho_r = 0.772$ g/ml
- Molar Specific heat ($40 - 80$°C): $C_{r,mol} = 119.6$ J/mol.K
- Mass Specific heat ($40 - 80$°C): $C_{r,mass} = 2824$ J/kg.K
- Latent heat of vaporization (86.5°C): $r_r = 39741$ J/mol

Water (H_2O):

- Molecular weight: $M_n = 18$ g/mol
- Specific weight ($40 - 80$°C): $\rho_n = 0.992$ g/ml
- Molar Specific heat ($40 - 80$°C): $C_{n,mol} = 75.24$ J/mol.K
- Mass Specific heat ($40 - 80$°C): $C_{n,mass} = 4186$ J/kg.K
- Latent heat of vaporization (86.5°C): $r_n = 41200$ J/mol

6.2 Property Calculation Formulas

The following parameters were calculated for each experiment:

1. **Flow rate conversion (ml/min):**

$$F_{ml/min} = F_{raw} \cdot 5.64 \tag{15}$$

2. **Molar fraction (x_D, x_F):**

$$x = \frac{C/M_r}{C/M_r + (1 - C)/M_n} \tag{16}$$

where C is the mass concentration.

3. **Molar flow rate (mol/min):**

$$n = \frac{V \cdot \rho}{M_{mix} \cdot 60} \tag{17}$$

4. **Average specific heat of the mixture (C_{mix}):**

$$C_{mix} = x \cdot C_R + (1 - x) \cdot C_N \tag{18}$$

5. **Average latent heat of the mixture (r_{mix}):**

$$r_{mix} = x \cdot r_R + (1 - x) \cdot r_N \tag{19}$$

6. **Feed quality (q):**

$$q = \frac{H_{GF} - H_F}{H_{GF} - H_{LF}} \tag{20}$$

7. **Overall Efficiency (E_0):**

$$E_0 = \frac{n_{LT}}{n_T} \tag{21}$$

where n_{LT} is the number of theoretical trays and n_T is the number of actual trays (5).

6.3 Supplemental Data Tables

Additional density values used for interpolation:

Table 5: Measured density values for various streams

Stream	**Temp (°C)**	ρ_r **(kg/m³)**	ρ_n **(kg/m³)**	ρ_{tb} **(kg/m³)**
Distillate (High)	100.0	716.20	960.60	-
Distillate (Low)	44.0	769.23	990.06	901.73
Feed	50.0	763.85	987.80	947.49
Reflux	38.0	774.542	992.096	905.33

6.4 Specific and Latent Heat Calculation

Table 6: Specific and latent heat values (average)

Parameter	**Value**
Boiling point $t_{F,si}$	89 °C
Molecular fraction x_F	0.0623
Specific Heat C_R (Ethanol)	3333.60 J/kg.K
Specific Heat C_N (Water)	4227.86 J/kg.K
Average Specific Heat C_F	4102.66 J/kg.K
Latent Heat r_F	2178129.90 J/kg.K

6.5 Nomenclature

Table 7: Nomenclature and Symbols

Symbol	**Description**	**Unit**
α	Relative volatility	[-]
x_D, x_F, x_W	Mole fraction of Ethanol (Distillate, Feed, Bottoms)	[-]
y_n	Mole fraction of Ethanol in vapor phase (stage n)	[-]
R	Reflux ratio	[-]
q	Feed condition parameter	[-]
L, V, D	Flow rates (Liquid, Vapor, Distillate)	[mol/ph]
ρ_r, ρ_n	Density (Ethanol, Water)	[kg/m^3]
r_r, r_n	Latent heat of vaporization	[J/mol]
E_0	Overall column efficiency	[-]
n_{LT}	Number of theoretical stages	[-]

7 REFERENCES

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